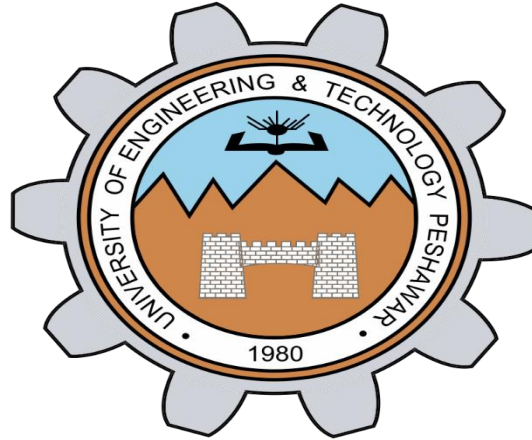


UNIVERSITY OF ENGINEERING AND TECHNOLOGY,
PESHAWAR PAKISTAN

Main Campus



Operating system lab mid exam

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Section E

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DEPARTMENT OF COMPUTER SCIENCE

UNIVERSITY OF ENGINEERING AND TECHNOLOGY, PESHAWAR, PAKISTAN

Q1: User Account Management

a) Create user

```
Session Actions Edit View Help
(kali@kali)-[~]
$ adduser labintern
fatal: Only root may add a user or group to the system.

(kali@kali)-[~]
$ sudo adduser labintern
[sudo] password for kali:
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for labintern
Enter the new value, or press ENTER for the default
  Full Name []: hamza
    Room Number []:
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n] y

(kali@kali)-[~]
$
```

b) Assign password

```
    Work Phone []:
    Home Phone []:
      Other []:
Is the information correct? [Y/n] y

(kali@kali)-[~]
$ whoami
kali

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ sudo passwd labinter
passwd: user 'labinter' does not exist

(kali@kali)-[~]
$ sudo passwd labintern
New password:
Retype new password:
passwd: password updated successfully

(kali@kali)-[~]
$
```

```
(kali@kali)-[~]
$ tail -n 1 /etc/passwd
labintern:x:1008:1008:hamza,,,:/home/labintern:/bin/bash

(kali@kali)-[~]
$
```

c) Verify user exists

```
(kali@kali)-[~]  
$ su - labintern  
Password:  
(labintern@kali)-[~]  
$
```

e) Display current username

```
(labintern@kali)-[~]  
$ whoami  
labintern
```

Log in as labintern

Q2: Directory and File Management

f) Create directory

```
(labintern@kali)-[~]  
$ mkdir ~/intern-project  
(labintern@kali)-[~]  
$ cd ~/intern-project  
(labintern@kali)-[~/intern-project]  
$ pwd  
/home/labintern/intern-project  
(labintern@kali)-[~/intern-project]  
$
```

g) Navigate & create file

```

(labintern@kali)-[~/intern-project]
$ pwd
/home/labintern/intern-project

(labintern@kali)-[~/intern-project]
$ ls -l
total 4
-rw-rw-r-- 1 labintern labintern 0 Nov 14 12:14 solution.c

(labintern@kali)-[~/intern-project]
$ ls -li
total 4
-rw-rw-r-- 1 labintern labintern 0 Nov 14 12:14 solution.c

(labintern@kali)-[~/intern-project]
$

```

h) Print absolute path

```

(labintern@kali)-[~/intern-project]
$ chomd 700 solution.c
Command 'chomd' not found, did you mean:
  command 'chmod' from deb coreutils
Try: apt install <deb name>

```

Its show error now we can update this

```

(kali@kali)-[~]
$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.1 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.6 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [118 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [276 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [185 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [905 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [14.3 kB]
Fetched 76.2 MB in 45s (1,702 kB/s)
997 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

```

(kali@kali)-[~]
$ cd intern-project

(kali@kali)-[~/intern-project]
$ touch solution.c

(kali@kali)-[~/intern-project]
$ pwd
/home/kali/intern-project

(kali@kali)-[~/intern-project]
$ ls -l
total 4
-rw-rw-r-- 1 kali kali 0 Nov 14 12:14 solution.c

(kali@kali)-[~/intern-project]
$

```

i) List detailed files

☑ Q3: File Permissions and Security

j) Set permissions (owner only full access)

k) Verify permissions

```
(kali㉿kali)-[~/intern-project]
$ chmod 700 solution.c

(kali㉿kali)-[~/intern-project]
$ ls -l solution.c
ls-1: command not found

(kali㉿kali)-[~/intern-project]
$ ls -l solution.c
solution.c

(kali㉿kali)-[~/intern-project]
$
```

Q4: Process Monitoring and Management

n) Show all running processes

```
(kali㉿kali)-[~/intern-project]
$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root            1  0.2  0.3 24852 15772 ?        Ss   01:59   0:04 /sbin/init splash
root            2  0.0  0.0      0     0 ?        S    01:59   0:00 [kthreadd]
root           33  0.0  0.0      0     0 ?        S    01:59   0:00 [pool_workqueue_release]
root           34  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/R-rcu_gp]
root           35  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/R-sync_wq]
root           36  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/R-kvfree_rcu_reclaim]
root           37  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/R-slub_flushwq]
root           38  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/R-netns]
root           39  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/0:0H-kblockd]
root           40  0.0  0.0      0     0 ?        I    01:59   0:00 [kworker/u16:0-ipv6_addrconf]
root           41  0.0  0.0      0     0 ?        I<   01:59   0:00 [kworker/R-mm_percpu_wq]
root           42  0.0  0.0      0     0 ?        S    01:59   0:00 [ksoftirqd/0]
root           43  0.1  0.0      0     0 ?        I    01:59   0:03 [rcu_preempt]
root           44  0.0  0.0      0     0 ?        S    01:59   0:00 [rcu_exp_par_gp_kthread_worker/0]
root           45  0.0  0.0      0     0 ?        S    01:59   0:00 [rcu_exp_gp_kthread_worker]
root           46  0.0  0.0      0     0 ?        S    01:59   0:00 [migration/0]
root           47  0.0  0.0      0     0 ?        S    01:59   0:00 [idle_inject/0]
root           48  0.0  0.0      0     0 ?        S    01:59   0:00 [cpuhp/0]
root           49  0.0  0.0      0     0 ?        S    01:59   0:00 [cpuhp/1]
```

```
(kali@kali)~/intern-project
$ ps -u labintern
PID TTY      TIME CMD
(kali@kali)~/intern-project
$
```

```
(kali@kali)~/intern-project
$ echo $$
1841
(kali@kali)~/intern-project
$
```

```
(kali@kali)~/intern-project
$ ps -o ppid= -p $$
1793
(kali@kali)~/intern-project
$
```

q) Run top

```
(kali㉿kali)-[~/intern-project]
$ ps -f
UID          PID    PPID  C  STIME TTY          TIME CMD
kali         1841    1793  0   02:01 pts/0        00:00:08 /usr/bin/zsh
kali         22782   1841  99   02:42 pts/0        00:00:00 ps -f

(kali㉿kali)-[~/intern-project]
$
```

u) Process States

State	Definition
R (Running)	Process is currently executing on CPU
S (Sleeping)	Waiting for an event or resource
Z (Zombie)	Process finished but still in process table
T (Stopped)	Process has been stopped (e.g., by signal)
D (Uninterruptible Sleep)	Waiting on I/O, cannot be interrupted